

Metheia

Travel Agency Web Application Software Requirements Specification 3.0

Prepared by

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1. Introduction

1.1 Purpose

The functional and non-functional requirements for Metheia, an online application that suggests trip destinations, are outlined in this Software Requirements Specification document. With dynamic pages and filtering, users can easily book, update, cancel, and pay for restaurants, hotels, and guides. This application shows comprehensive travel details, including well-known sights, lodging, dining options, local guides, transportation choices, cultural insights, and estimated budgets that are dynamically retrieved from a database.

1.2 Scope

Metheia includes the design, development, testing, and deployment of a full-stack web application that:

- Presents an interactive chatbox to connect with admin
- Displays comprehensive destination details, including attractions, restaurants, hotels and transportation.
- Enables search and filtering of destinations, hotels, restaurants and guides..
- Supports restaurant and guide booking features.
- Allows users to provide feedback for future improvements.

1.4 Target Users

- For users of all ages, genders, locations, and occupations.
- For users with an interest in travelling.

3. System Requirements

3.1 Functional Requirements

Homepage

- FR-1: Display homepage with featured destinations and offers.
- FR-2: Search destinations.
- FR-3: Allow users to go to other pages through navbar.
- FR-4: Allow users to login or sign up.

Destination & Information Display

- FR-5: Retrieve and display destination details, including description, and key facts.
- FR-6: Display top tourist attractions and local restaurants.
- FR-7: Show available transport modes and average costs.
- FR-8: Present hotel listings with filters (price, rating, etc.).
- FR-9: Show weather, travel seasons, and budget insights.
- FR-10: Display local language, currency, and cultural etiquette.
- FR-11: Provide packing checklist and safety tips.
- FR-12: Allow logged in users to give like or dislike.
- FR-13: Allow logged in users to leave comments.

Profile

- FR-14: Allow logged in users to view and edit name, email and password.
- FR-15: Allow logged in users to delete account.

Hotels

- FR-16: Support secure payment transactions.
- FR-17: Hotel listings with filters (price, rating, location, amenities).
- FR-18: Real-time availability check and booking confirmation.
- FR-19: Hotel booking.
- FR-20: Modifying hotel booking information (Edit, Cancel).

Restaurant

- FR21: Allow users to search and filter restaurants by name, location, price range, and cuisine, requiring at least one active filter.
- FR22: Enable users to book tables by providing personal details, validating seat availability, and confirming bookings.
- FR23: Display complete restaurant details, including address, cuisine, price range, and available seats before booking.
- FR24: Allow users to view booking history using their email with full booking details.
- FR25: Provide filter management with enable options, a Clear All Filters function.

Guides

- FR-26: Filter guides by name
- FR-27: Booking guides
- FR-28: Show booking confirmation

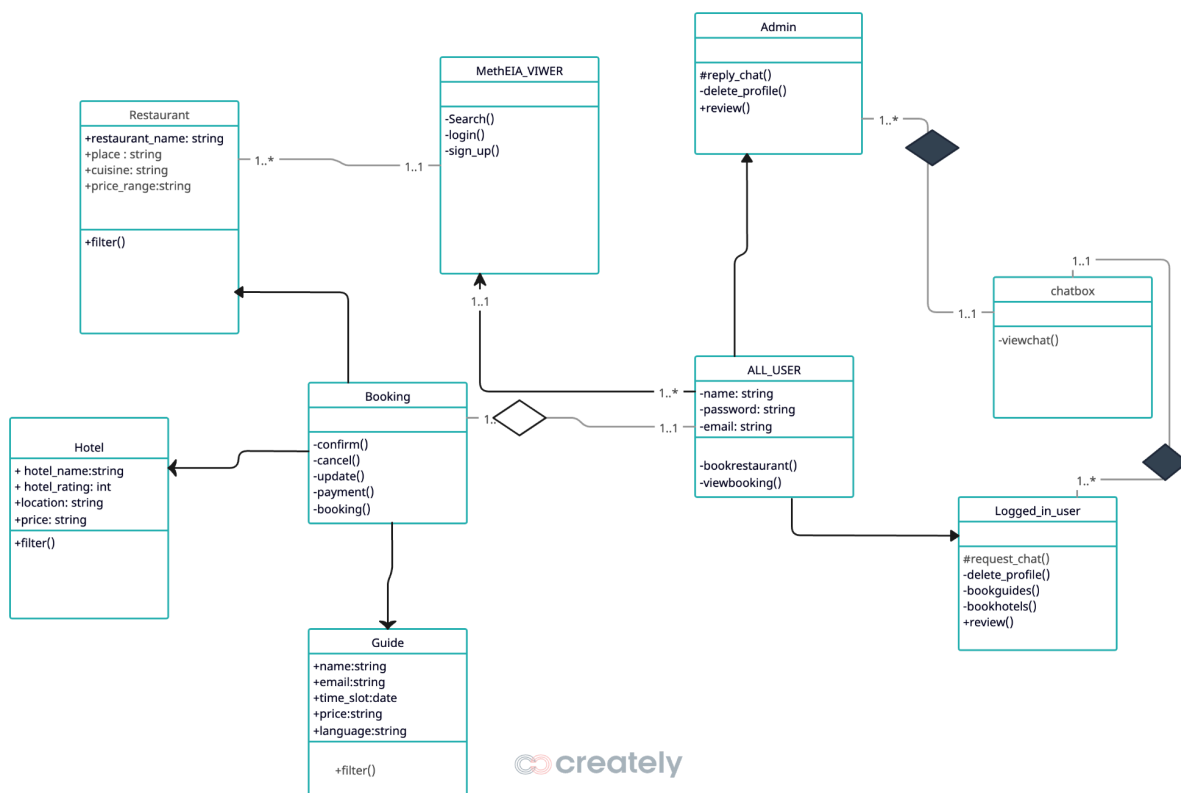
Chat

- FR29: Allow users to send messages and receive immediate visual feedback.
- FR30: Route messages to administrators and allow admins to respond while updating message status.
- FR31: Persist and display complete chat history in chronological order for each user.
- FR32: Provide a globally accessible chatbox interface with an open toggle.

3.2 Non-Functional Requirements

- Must return responses within 2 seconds under normal load.
- Must be simple to use and visually appealing.
- The interface shall be responsive, intuitive, and usable across desktop, tablet, and mobile devices.

4. Class diagram



5. Technology Stack

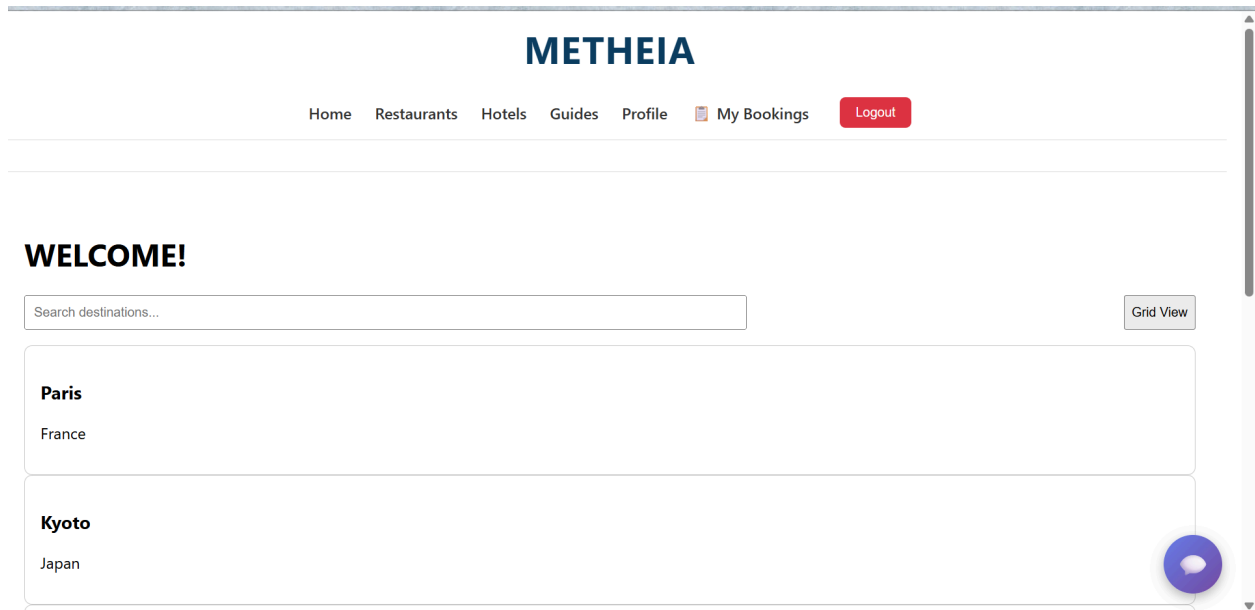
5.1 MERN Stack Components

- **MongoDB:** Stores destinations, hotels, restaurants, transportations, etc.
- **Express.js:** Handles routing
- **React.js:** Frontend for dynamic search, and results display.
- **Node.js:** Backend runtime

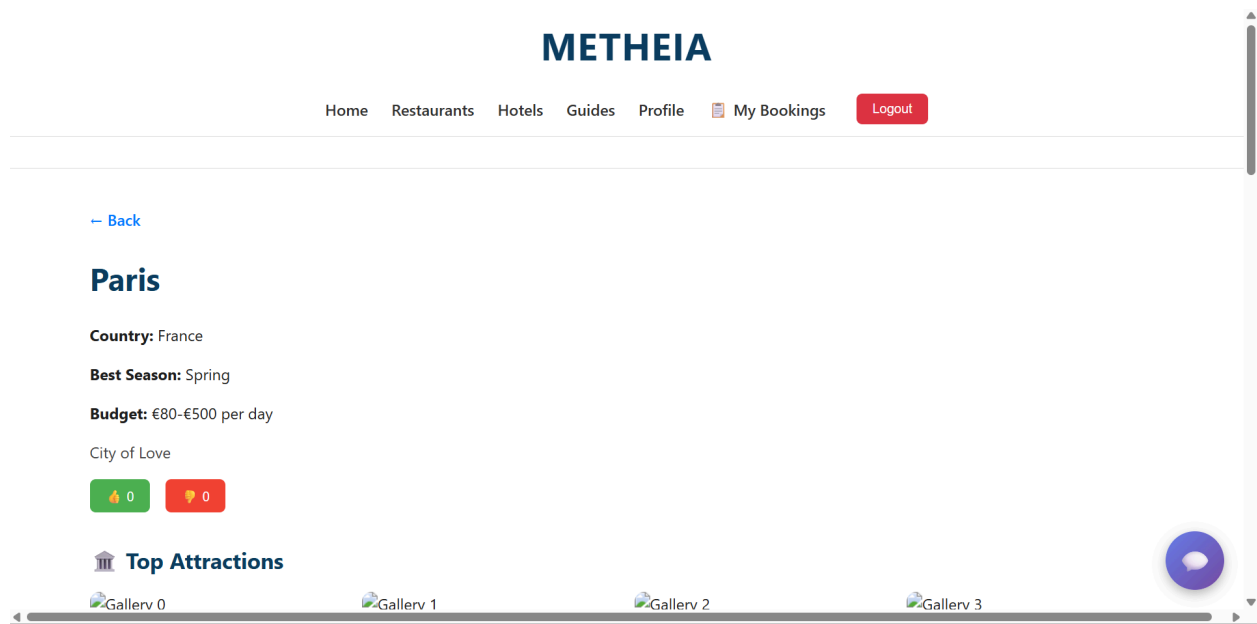
6. Compatibility with system environment

Metheia, currently configured for a **Localhost Environment**, is built using the MERN stack, ensuring high compatibility across diverse system environments. On the **server-side**, the application is compatible with any environment supporting Node.js. The backend is powered by Express.js, requiring a local runtime environment. On the **client-side**, the application is built with React.js and is fully responsive and compatible with all modern web browsers (Chrome, Firefox, Safari, Edge). The application works whether the user is accessing it from Windows or macOS. Furthermore, the use of **Environment Variables (.env)** ensures that the system can easily transition between development, testing, and production environments without code modification.

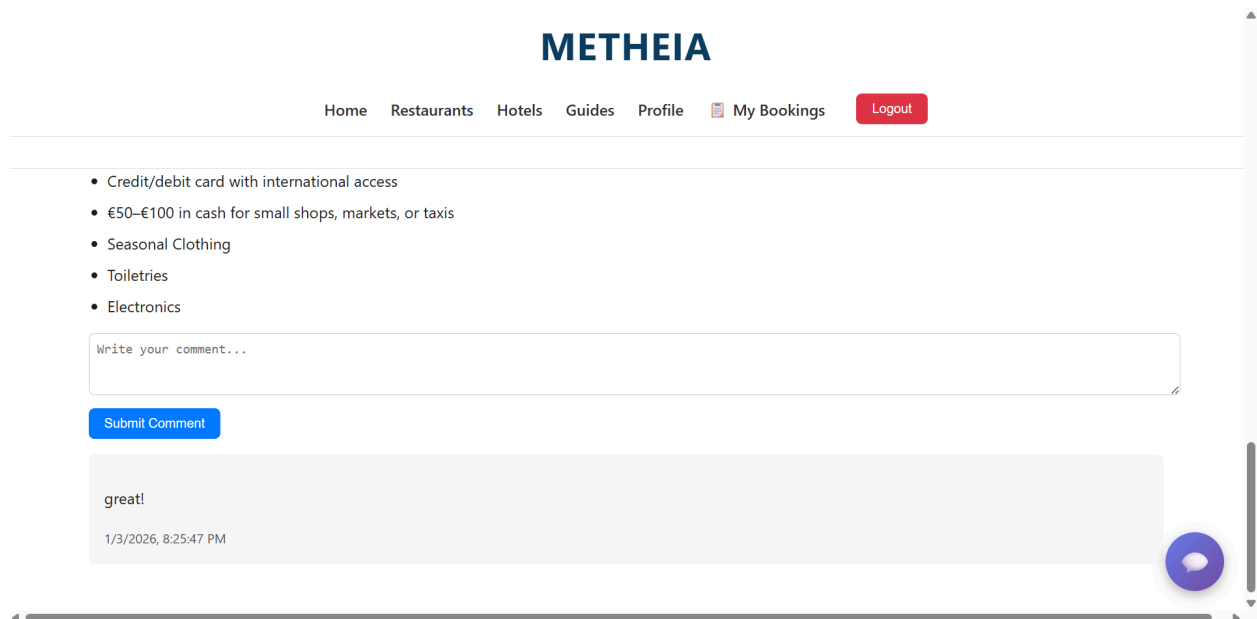
7. Implementation



The homepage, where we can view destinations and search in the search bar. The chatbox can be used to connect to the admin.



Destination page, where users can view destination data and give like or dislike.



Users can leave comments.

METHEIA

[Home](#) [Restaurants](#) [Hotels](#) [Guides](#) [Profile](#) [My Bookings](#)

[Logout](#)

Select Filter Options (Choose at least one)

You can select multiple filters to refine your search

☐ Restaurant Name

☒ Place/Destination

☐ Price Range

☐ Cuisine Type

london

 Search Restaurants

Clear All Filters

2 Restaurants Found

Filter restaurants.

METHEIA

[Home](#) [Restaurants](#) [Hotels](#) [Guides](#) [Profile](#) [My Bookings](#)

[Logout](#)



Find Your Perfect Restaurant

Select at least one filter option and search to discover amazing dining experiences

[View My Bookings](#)

Select Filter Options (Choose at least one)

You can select multiple filters to refine your search

☐ Restaurant Name

☒ Place/Destination

☐ Price Range

☐ Cuisine Type

london

[Search Restaurants](#)

[Clear All Filters](#)

2 Restaurants Found



Dishoom Covent Garden

★★★★★ 4.7

Location:

London, United Kingdom

Price Range:

£ 60 - 500

Available Seats:

9

Contact Number:

+44 20 7420 9320

Cuisine:

Indian Fusion

Best Dishes:

- Chicken Ruby
- House Black Daal
- Bacon Naan Roll

[Visit Website](#)

[Pre-Book Table](#)



Rules Restaurant

★★★★★ 4.6

Location:

London, United Kingdom

Price Range:

£ 60-200

Available Seats:

9

Contact Number:

+44 20 7836 5314

Cuisine:

British European

Best Dishes:

- Steak and Kidney Pie
- Roast Duck
- Sticky Toffee Pudding

[Visit Website](#)

[Pre-Book Table](#)

View restaurant data according to the filter.

METHEIA

[Home](#) [Restaurants](#) [Hotels](#) [Guides](#) [Profile](#) [My Bookings](#)

[Logout](#)



Find Your Perfect Restaurant

Select at least one filter option and search to discover amazing dining experiences

[Search Restaurants](#)

My Restaurant Bookings

s@gmail.com

[View Bookings](#)

Rules Restaurant

Pending

Date: 2026-01-15

Time: 22:27

Guests: 1

Name: rtyui

Email: s@gmail.com

Phone: 23456

Booked On: 1/3/2026, 10:25:02 PM

Dishoom Covent Garden

Cancelled

Date: 2026-01-03

Time: 21:00

Guests: 1

Name: asdfghujkiol;

Email: s@gmail.com

Phone: 23456

Special Requests: dfghj

Booked On: 1/3/2026, 9:58:25 PM

[View booked restaurants.](#)

Pre-Book Table at Rules Restaurant



Full Name *

Enter your full name

Email *

Enter your email

Phone Number

Enter your phone number

Date *

mm/dd/yyyy



Time *

--:-- --



Number of Guests *

1

Restaurant booking form.

METHEIA

Available Hotels

Filters

Location:

Min Price (\$):

Max Price (\$):

Min Rating:

Amenities:

Beach Access

☐

Breakfast

☐

Fireplace

☐

Gym

☐

METHEIA

Restaurant

Ski Access

☐

Spa

☐

WiFi

☐

Apply Filters

Reset Filters

Urban Stay Hotel

Tokyo, Japan

\$270/night

4.6 stars

Amenities:

WiFiGymRestaurant

View Details

City Lights Hotel

Chicago, USA

\$180/night

4.2 stars

Amenities:

WiFiRestaurantParking

Hotels filter

METHEIA

[Home](#) [Restaurants](#) [Hotels](#) [Guides](#) [Profile](#) [My Bookings](#)

[Logout](#)

Urban Stay Hotel

 Tokyo, Japan

Price per Night

\$270

Rating

 4.6 stars

 0

 0

Comments

Write your comment...

[Submit Comment](#)

No comments yet.

[Book Hotel](#)

[Back to Hotels](#)

Hotel page with data, like or dislike, comment and booking options.

Select a Guide

Kyoto

▼

Filter by name

Yuki Tanaka - ¥5000/hr

Hiroshi Yamamoto - ¥4500/hr

Filter guides.

Kyoto 

Filter by name

Yuki Tanaka - ¥5000/hr

Hiroshi Yamamoto - ¥4500/hr

Book Yuki Tanaka

san

01/08/2026 

12:06 PM 

12:06 AM 

Duration: 12.0 hours

Rate: \$5000.00/hour

Total Cost: \$60000.00

Book Now

Booking guide.



My Profile

Name:

Update Name

Email:

Update Email

New Password:

Update Password



Danger Zone

Once you delete your account, there is no going back. Please be certain.



Delete My Account

View and edit user info. Delete account.

My Bookings

Hotel Bookings (3 active, 1 cancelled)

Edit Booking - Grand Palace Hotel

Check-in Date

01/15/2026

Check-out Date

01/23/2026

Guests

1

Save

Cancel

Urban Stay Hotel

Tokyo, Japan

Confirmed

Check-in: 2/4/2026

Check-out: 3/12/2026

Guests: 1

Total: \$9720

Edit

Cancel Booking

Ocean View Resort

Malibu, USA

Confirmed

Restaurant Bookings (1 active, 1 cancelled)

Rules Restaurant

pending

Date: 2026-01-15

Time: 22:27

Guests: 1

Edit

Cancel Booking

Guide Bookings (2 active, 1 cancelled)

Guide Booking

Bali

confirmed

Date: 1/29/2026

0

Cost: \$\$NaN

Edit

Pay Now

Cancel Booking

Guide Booking

Kyoto

pending

Date: 1/9/2026

Hours: 12

Cost: \$\$NaN

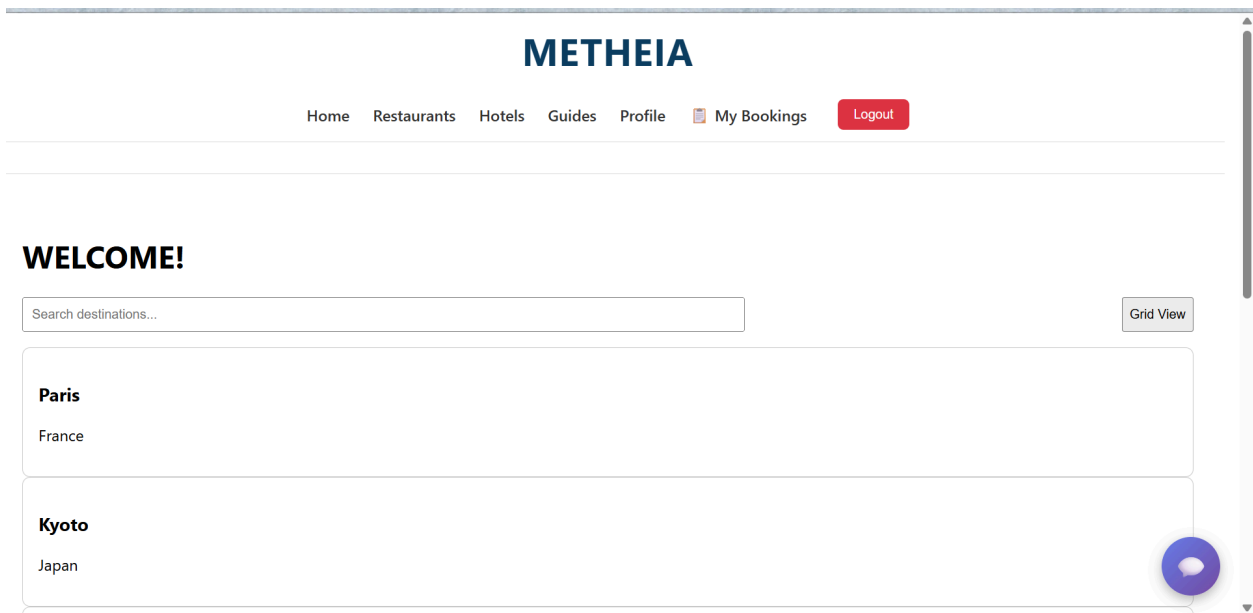
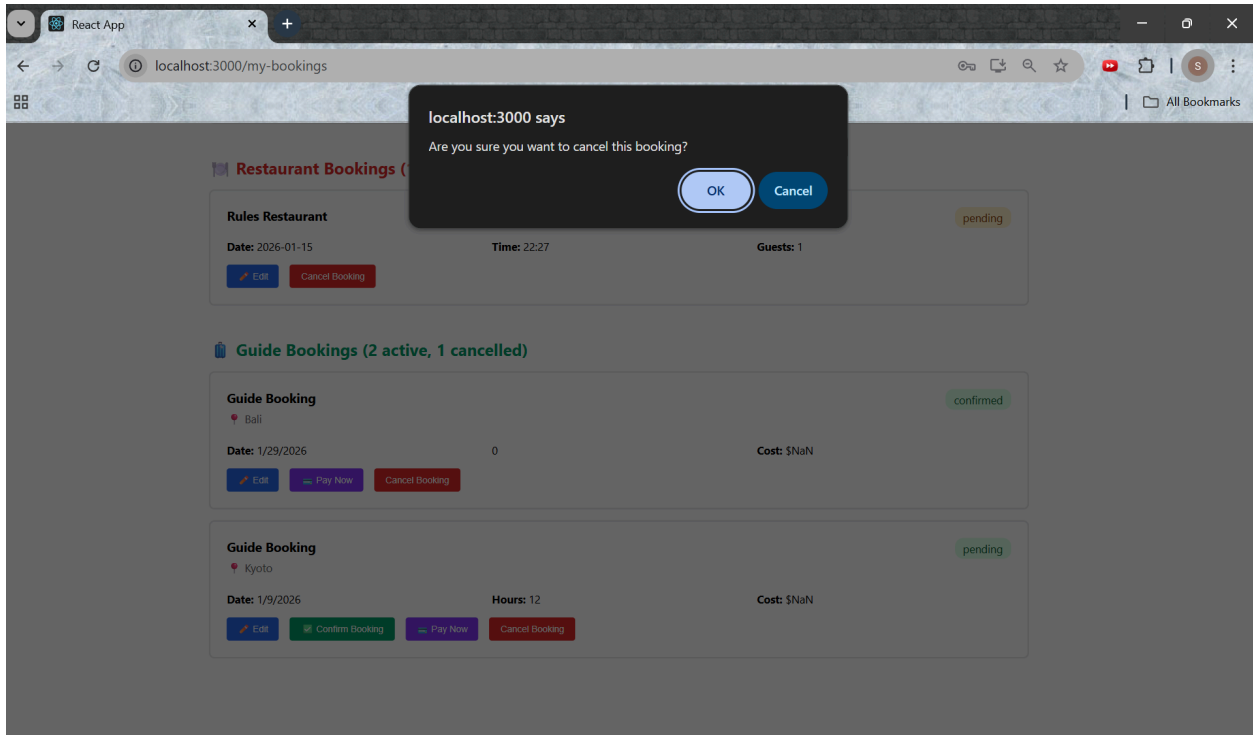
Edit

Confirm Booking

Pay Now

Cancel Booking

View, edit, and delete all user bookings.



Cancel booking.



Payment for Guide Booking

Booking Details

Destination: Rome

Hours: 12

Date: 1/7/2026

Total Amount: €660.00

Card Number

1234 5678 9012 3456

Cardholder Name

John Doe

Expiry Date (MM/YY)

12/25

CVV

123

Pay €660.00

Cancel

Payment for hotel booking.

8. Challenges

- Booking payment does not work with a Visa card.
- Images do not load.
- Admin functions not complete.
- It took some days to integrate the entire system together.

9. Conclusion

Metheia is a MERN-based travel destination web application that connects frontend and backend layers and uses MongoDB collections to deliver personalized, data-driven travel

experiences. It provides a responsive and user-friendly interface, ensuring smooth interaction, scalability, and easy database integration, ideal for a modern travel agency solution.

10. References

MERN Stack Documentation

- [MongoDB](#)
- [Express.js](#)
- [React.js](#)
- [Node.js](#)
- [Agile](#)